

Table of contents

Exam questions	2
Relationship between Linear Discriminant Analysis (LDA) and Conditional Modeling for Classification	2
Question	2
Introduction to Conditional Modeling in Classification	2
Bayesian Framework for Classification	3
Bayes' Rule for Optimal Decision	3
LDA as a Conditional Generative Model	3
Fundamental Assumptions of LDA	3
Explicit Formulation of the LDA Model	4
From Conditional Modeling to Linear Discriminant Function	4
Derivation of the Decision Function	4
Explicit Form of the Decision Boundary	4
Geometric Interpretation and Link with Fisher's Criterion	5
Optimal Projection for Discrimination	5
Equivalence with the Probabilistic Approach	5
Model Parameter Estimation	5
Maximum Likelihood Approach	5
Advantages of the Conditional Interpretation of LDA	6
1. Solid Probabilistic Foundations	6
2. Ability to Handle Imbalanced Classes	6
3. Extension to Multi-Class Classification	6
4. Interpretation in Terms of Likelihood Ratio	6
Limitations and Extensions	7
Restrictive Common Covariance Assumption	7
Modern Extensions	7
Concrete Example: Iris Flower Classification	7
Conclusion	7
Linear Discriminant Analysis (LDA): Comprehensive Explanation	8
Question	8
What is Linear Discriminant Analysis?	8
Fundamental Assumptions of LDA	8
Gaussian Distribution	9
Homoscedasticity (Common Covariance)	9
Independent Observations	9
Class Separability	9
Calculation and Methodology	9
Notations and Definitions	9
Dispersion Matrices	10

Fisher's Criterion	10
Optimization Solution	10
Detailed Calculation Steps	11
Interpretation of Results	11
Discriminant Dimensions	11
Explained Separation Percentage	12
Coefficient Significance	12
Visualization	12
Use for Classification	12
Probabilistic Modeling Approach	12
Probability Estimation	12
Linear Discriminant Functions	13
Decision Rule	13
Decision Boundaries	13
Classification Process for a New Observation	13
Parameter Estimation	14
Advantages and Limitations	14
Advantages	14
Limitations	14
Alternatives and Extensions	14
Concrete Example: Iris Flower Classification	15
Problem Context	15
LDA Application	15
Conclusion	15

Exam questions

Relationship between Linear Discriminant Analysis (LDA) and Conditional Modeling for Classification

Question

Explain how the model of Linear Discriminant Analysis (LDA) relate to conditional model for classification.

Introduction to Conditional Modeling in Classification

In supervised classification problems, we aim to predict the class label y of an observation x . A fundamental approach involves modeling the **conditional probability** $P(y|x)$, i.e., the probability that an observation x belongs to a given class y .

Bayesian Framework for Classification

Bayes' Rule for Optimal Decision

According to Bayes' theorem, the posterior probability that an observation x belongs to class k is:

$$P(C = k|x) = \frac{P(x|C = k) \cdot P(C = k)}{P(x)}$$

Where: - $P(C = k)$ is the **prior probability** of class k - $P(x|C = k)$ is the **likelihood** of x given class k - $P(x)$ is the marginal probability of x (constant for all classes)

The optimal decision rule (minimizing the error rate) consists of assigning x to the class k that maximizes the posterior probability:

$$\delta(x) = \arg \max_k P(C = k|x)$$

Since $P(x)$ is constant, this rule is equivalent to:

$$\delta(x) = \arg \max_k [P(x|C = k) \cdot P(C = k)]$$

LDA as a Conditional Generative Model

Fundamental Assumptions of LDA

Linear Discriminant Analysis is based on three clear assumptions:

1. **Gaussian distribution per class:** Each class follows a multivariate normal distribution

$$P(x|C = k) \sim \mathcal{N}(\mu_k, \Sigma)$$

2. **Common covariance matrix:** All classes share the **same** covariance matrix Σ
3. **Prior probabilities:** Classes have prior probabilities $P(C = k)$ (often estimated by sample proportions)

Explicit Formulation of the LDA Model

Under these assumptions, the probability density for class k is written as:

$$P(x|C = k) = \frac{1}{(2\pi)^{D/2}|\Sigma|^{1/2}} \exp\left(-\frac{1}{2}(x - \mu_k)^T \Sigma^{-1}(x - \mu_k)\right)$$

Where D is the data dimension, μ_k is the mean vector of class k , and Σ is the common covariance matrix.

From Conditional Modeling to Linear Discriminant Function

Derivation of the Decision Function

By applying the Bayesian decision rule and taking the logarithm (which preserves maxima), we obtain:

$$\begin{aligned}\delta(x) &= \arg \max_k [\ln P(x|C = k) + \ln P(C = k)] \\ &= \arg \max_k \left[-\frac{1}{2}(x - \mu_k)^T \Sigma^{-1}(x - \mu_k) + \ln P(C = k) + \text{constant} \right]\end{aligned}$$

By expanding the quadratic term and eliminating terms independent of k , we obtain the **linear discriminant function**:

$$\delta(x) = \arg \max_k \left[x^T \Sigma^{-1} \mu_k - \frac{1}{2} \mu_k^T \Sigma^{-1} \mu_k + \ln P(C = k) \right]$$

Explicit Form of the Decision Boundary

The discriminant function for class k can be written as a linear function in x :

$$g_k(x) = w_k^T x + b_k$$

Where: - $w_k = \Sigma^{-1} \mu_k$ is the weight vector for class k - $b_k = -\frac{1}{2} \mu_k^T \Sigma^{-1} \mu_k + \ln P(C = k)$ is the bias for class k

The decision boundary between two classes k and l is then **linear**:

$$\{x : g_k(x) = g_l(x)\} \quad \Rightarrow \quad (w_k - w_l)^T x + (b_k - b_l) = 0$$

Geometric Interpretation and Link with Fisher's Criterion

Optimal Projection for Discrimination

LDA seeks the projection direction a that maximizes the relative separation between classes. Fisher's criterion, which maximizes:

$$J(a) = \frac{a^T \Sigma_b a}{a^T \Sigma_w a}$$

corresponds exactly to finding the direction that maximizes the ratio between: - Σ_b : between-class dispersion (to maximize) - Σ_w : within-class dispersion (to minimize)

Equivalence with the Probabilistic Approach

When the prior probabilities are equal ($P(C = k) = 1/M$), the LDA solution via Fisher's criterion is equivalent to the solution obtained by the Gaussian modeling approach with common covariance.

The optimal direction a is given by the first eigenvector of $\Sigma_w^{-1} \Sigma_b$, which in the two-class case reduces to:

$$a \propto \Sigma^{-1}(\mu_1 - \mu_2)$$

Exactly the same expression as in the Bayesian approach!

Model Parameter Estimation

Maximum Likelihood Approach

From a training sample $\{(x_i, y_i)\}_{i=1}^N$, LDA parameters are naturally estimated:

- **Class means:** $\hat{\mu}_k = \frac{1}{N_k} \sum_{i: y_i=k} x_i$
- **Common covariance matrix:** $\hat{\Sigma} = \frac{1}{N-M} \sum_{k=1}^M \sum_{i: y_i=k} (x_i - \hat{\mu}_k)(x_i - \hat{\mu}_k)^T$
- **Prior probabilities:** $\hat{P}(C = k) = \frac{N_k}{N}$

These estimators are **consistent** and **asymptotically efficient** under the model assumptions.

Advantages of the Conditional Interpretation of LDA

1. Solid Probabilistic Foundations

Unlike a simple search for discriminant directions, LDA provides a **complete generative model** that allows: - Generating synthetic data for each class - Calculating well-calibrated posterior probabilities - Incorporating prior knowledge about classes

2. Ability to Handle Imbalanced Classes

By using prior probabilities $P(C = k)$, LDA can naturally account for unequal class frequencies.

3. Extension to Multi-Class Classification

The probabilistic framework naturally extends to $M > 2$ classes, whereas Fisher's criterion requires generalizations.

4. Interpretation in Terms of Likelihood Ratio

The decision between two classes k and l can be expressed as:

$$\log \frac{P(C = k|x)}{P(C = l|x)} = (w_k - w_l)^T x + (b_k - b_l)$$

This corresponds to a **likelihood ratio test** with a threshold adapted to prior probabilities.

Limitations and Extensions

Restrictive Common Covariance Assumption

The main limitation of LDA is the assumption that all classes share the same covariance matrix Σ .

When this assumption is not verified, we use: - **QDA (Quadratic Discriminant Analysis)**: allows different covariances per class, leading to quadratic boundaries - **Regularizations**: regularized versions of LDA for high-dimensional data

Modern Extensions

- **Flexible LDA**: uses kernels to capture non-linearities
 - **Sparse LDA**: introduces sparsity constraints for variable selection
 - **Robust LDA**: versions resistant to outliers
-

Concrete Example: Iris Flower Classification

Consider the Iris dataset with 3 flower species.

LDA models each species by a Gaussian distribution in the feature space (petal/sepal length and width).

For a new flower measurement x , LDA calculates: 1. The likelihood $P(x|\text{setosa})$, $P(x|\text{versicolor})$, $P(x|\text{virginica})$ 2. The posterior probabilities via Bayes' rule 3. Assigns the species with the maximum posterior probability

The boundary between species appears linear in the feature space, due to the common covariance assumption.

Conclusion

Linear Discriminant Analysis is **fundamentally a conditional model for classification** that:

1. **Explicitly models** $P(x|C = k)$ as Gaussians with common covariance
2. **Uses Bayes' rule** to calculate $P(C = k|x)$ and make decisions
3. **Reduces to linear discriminant functions** thanks to its simplifying assumptions

4. **Is equivalent to Fisher's criterion** under the assumption of balanced classes

This dual perspective - both as a probabilistic generative model and as a discriminant projection method - explains the power and enduring popularity of LDA in statistical learning.

LDA perfectly illustrates how **well-chosen probabilistic assumptions** can lead to **simple and interpretable algorithms** while remaining theoretically grounded and practically effective.

Linear Discriminant Analysis (LDA): Comprehensive Explanation

Question

Explain the method of Linear Discriminant Analysis (LDA) including assumptions, calculation, and interpretation of the technique.

How is this technique used for classification?

What is Linear Discriminant Analysis?

Linear Discriminant Analysis (LDA) is a supervised statistical method that serves two main functions: **classification** and **dimensionality reduction**.

It aims to find linear combinations of the original variables that best separate different predefined classes.

Fundamental Assumptions of LDA

For LDA to function optimally, it relies on several key assumptions:

Gaussian Distribution

Each class is assumed to follow a multivariate Gaussian distribution. Formally, for each class k , observations are drawn from:

$$P(x|C = k) \sim \mathcal{N}(\mu_k, \Sigma)$$

where μ_k is the mean vector of class k and Σ is the covariance matrix.

Homoscedasticity (Common Covariance)

All classes share the same covariance matrix Σ . This assumption is crucial as it guarantees that decision boundaries will be linear.

Independent Observations

Observations are sampled independently and identically distributed within each class.

Class Separability

Class centers (μ_k) must be sufficiently distinct relative to within-class dispersion.

Calculation and Methodology

Notations and Definitions

Consider a problem with K classes:

- N_k : number of observations in class k
- N : total number of observations, with $N = \sum_{k=1}^K N_k$
- μ_k : mean vector of class k (dimension D)
- μ : global mean vector of all observations

Dispersion Matrices

Within-Class Scatter Matrix

This matrix measures variability within each class:

$$\Sigma_w = \sum_{k=1}^K \sum_{x_i \in \text{class } k} (x_i - \mu_k)(x_i - \mu_k)^T$$

Between-Class Scatter Matrix

This matrix measures dispersion between centers of different classes:

$$\Sigma_b = \sum_{k=1}^K N_k (\mu_k - \mu)(\mu_k - \mu)^T$$

Fisher's Criterion

LDA's objective is to find a projection vector a that maximizes the relative separation between classes.

Fisher's criterion formalizes this objective:

$$J(a) = \frac{a^T \Sigma_b a}{a^T \Sigma_w a}$$

The numerator $a^T \Sigma_b a$ represents the projected between-class dispersion (to maximize), while the denominator $a^T \Sigma_w a$ represents the projected within-class dispersion (to minimize).

Optimization Solution

Maximizing $J(a)$ reduces to solving a generalized eigenvalue problem:

$$\Sigma_b a = \lambda \Sigma_w a$$

In practice, we compute the eigenvectors of $\Sigma_w^{-1} \Sigma_b$ (provided Σ_w is invertible).

Detailed Calculation Steps

Step 1: Calculate Descriptive Statistics

- Class means: $\hat{\mu}_k = \frac{1}{N_k} \sum_{i:y_i=k} x_i$
- Global mean: $\hat{\mu} = \frac{1}{N} \sum_{i=1}^N x_i$

Step 2: Calculate Dispersion Matrices

- Within-class matrix: $\hat{\Sigma}_w = \sum_{k=1}^K \sum_{i:y_i=k} (x_i - \hat{\mu}_k)(x_i - \hat{\mu}_k)^T$
- Between-class matrix: $\hat{\Sigma}_b = \sum_{k=1}^K N_k (\hat{\mu}_k - \hat{\mu})(\hat{\mu}_k - \hat{\mu})^T$

Step 3: Solve Eigenvalue Problem

- Compute matrix $\hat{\Sigma}_w^{-1} \hat{\Sigma}_b$
- Extract eigenvectors corresponding to the largest eigenvalues

Step 4: Form Discriminant Space

- Sort eigenvectors by descending eigenvalues
- Form projection matrix $W = [a_1, a_2, \dots, a_{K-1}]$ (at most $K - 1$ non-zero directions)

Step 5: Project Data

$$Y = XW$$

where Y represents the data projected into the new discriminant space.

Interpretation of Results

Discriminant Dimensions

Each eigenvector a_i corresponds to a direction in the original space along which class separation is maximal.

The associated eigenvalue λ_i measures the discriminant power of this direction.

Explained Separation Percentage

We can evaluate the relative importance of each discriminant dimension:

$$\text{Proportion}_i = \frac{\lambda_i}{\sum_{j=1}^{K-1} \lambda_j} \times 100\%$$

Coefficient Significance

The coefficients of vector a_i indicate the relative importance of each original variable in discrimination.

A high absolute coefficient means the variable strongly contributes to class separation along that direction.

Visualization

By projecting the data onto the first two discriminant dimensions, we obtain a 2D visualization that allows appreciating the separation between classes.

Use for Classification

Probabilistic Modeling Approach

LDA can be interpreted as a generative model where each class is modeled by a normal distribution:

$$P(x|C = k) = \frac{1}{(2\pi)^{D/2} |\Sigma|^{1/2}} \exp \left(-\frac{1}{2} (x - \mu_k)^T \Sigma^{-1} (x - \mu_k) \right)$$

Probability Estimation

Prior Probabilities

Estimated by observed proportions in the sample:

$$\hat{P}(C = k) = \frac{N_k}{N}$$

Bayes' Rule for Classification

The probability that an observation x belongs to class k is given by:

$$P(C = k|x) = \frac{P(x|C = k)P(C = k)}{\sum_{l=1}^K P(x|C = l)P(C = l)}$$

Linear Discriminant Functions

For each class k , we define a linear function:

$$g_k(x) = w_k^T x + b_k$$

with: $w_k = \Sigma^{-1} \mu_k$ - $b_k = -\frac{1}{2} \mu_k^T \Sigma^{-1} \mu_k + \ln P(C = k)$

Decision Rule

We assign x to the class that maximizes the discriminant function:

$$\hat{y} = \arg \max_k g_k(x)$$

This is equivalent to choosing the class with the highest posterior probability $P(C = k|x)$.

Decision Boundaries

The boundary between two classes k and l is linear:

$$\{x : g_k(x) = g_l(x)\} \Rightarrow (w_k - w_l)^T x + (b_k - b_l) = 0$$

This linearity is a direct consequence of the common covariance assumption.

Classification Process for a New Observation

To classify a new observation x_{new} :

1. **Compute scores:** Evaluate $g_k(x_{\text{new}})$ for each class k
2. **Decision:** Assign to the class with the highest score
3. **Probabilistic option:** Calculate posterior probabilities $P(C = k|x_{\text{new}})$ for a probabilistic decision

Parameter Estimation

From a training sample $\{(x_i, y_i)\}_{i=1}^N$:

- Class means: $\hat{\mu}_k = \frac{1}{N_k} \sum_{i:y_i=k} x_i$
 - Common covariance: $\hat{\Sigma} = \frac{1}{N-K} \sum_{k=1}^K \sum_{i:y_i=k} (x_i - \hat{\mu}_k)(x_i - \hat{\mu}_k)^T$
 - Prior probabilities: $\hat{P}(C = k) = \frac{N_k}{N}$
-

Advantages and Limitations

Advantages

- **Solid theoretical foundations** based on Bayesian theory and Fisher's criterion
- **Computational efficiency**: analytical solution without iterative optimization
- **Interpretability**: discriminant directions have clear geometric meaning
- **Dual functionality**: integrated dimensionality reduction and classification
- **Robustness** to slight violations of normality

Limitations

- **Sensitivity to outliers** affecting mean and covariance estimates
- **Common covariance assumption** sometimes unrealistic in practice
- **Linearity**: cannot capture non-linear relationships between variables
- **Normality requirement**: degraded performance with strongly non-Gaussian distributions

Alternatives and Extensions

- **QDA (Quadratic Discriminant Analysis)**: relaxes common covariance assumption, giving quadratic boundaries
 - **Regularized LDA**: introduces regularization for high-dimensional problems
 - **Flexible Discriminant Analysis**: uses basis functions to capture non-linear relationships
 - **Support Vector Machines (SVM)**: alternative approach for linear and non-linear classification
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Concrete Example: Iris Flower Classification

Problem Context

The Iris dataset contains 3 flower species with 4 morphological characteristics: - Sepal length and width - Petal length and width

LDA Application

Step 1: Parameter estimation

Calculate means for each species and common covariance matrix.

Step 2: Discriminant direction calculation

Solve eigenvalue problem to obtain 2 directions (since $K - 1 = 2$).

Step 3: Projection and visualization

Reduce data to 2 dimensions, revealing good separation between species.

Step 4: Classification

Construct discriminant functions allowing classification of new flowers with high accuracy (>95%).

Conclusion

Linear Discriminant Analysis is an **elegant and powerful** method that combines several approaches:

- A **geometric perspective** seeking optimal projection directions
- A **probabilistic modeling** based on normal distributions
- An **optimal Bayesian decision rule** under its assumptions

Its **versatility** (classification and dimensionality reduction) and **high interpretability** make it a valuable tool in exploratory data analysis and predictive modeling.

However, its successful application depends on the reasonable fulfillment of its underlying assumptions.

When these assumptions hold, LDA provides an **efficient, interpretable, and theoretically grounded** solution to classification and group separation problems.